

Business Rules for the National Education Data Dictionary

Introduction

The National Education Data Dictionary is designed to support consistent, reliable, and efficient use of education data across jurisdictions, agencies, and systems. To guide its purpose and use, the following business rules describe the core functions and benefits of the data dictionary. These rules highlight its role as a knowledge base, a tool for interoperability, a foundation for data modelling, and an enabler of effective reporting, planning, and compliance.

Rules

- It is a **knowledge base**, collecting national agreements about data definitions into the one well-managed repository. This knowledge base can serve as a digital source of truth for schools, software developers and policy makers, and it can be consulted for work around data exchanges in the sector—including identifying where such data is already being collected.
- It is a national starting point for **education data modelling**. There has been interest among school authorities around creating data models at rest, as enterprise models for all the data that authorities need to manage and access, especially in data warehouse and business intelligence systems. A national data dictionary can help school authorities formulate their local data models, and the mappings necessary from existing data formats.
- It can be used to **enable interoperability between different formats** in data exchanges, and to mitigate discrepancies between data models. If the mapping between the target data formats via the data dictionary is precise, conversion between the two formats can be fully automated. Even if not, the data dictionary can still establish the subset of data that is fully interoperable: at least partial interoperability can be achieved between different data formats, even if they cannot be mapped completely. So the data dictionary can form the basis of fully automated, or at least semi-automated translation between different data standards.
- It can be used to **populate multiple data collections from the same source data**. The data dictionary identifies elements in common between different data collections, along with any business rules or definition modifications specific to each collection. Schools can use this to fulfil their data reporting obligations more efficiently, by consolidating the data gathering that recurs among different collection instances.
- It can be used to **plan for data reporting and extraction from multiple data sources**, and to automate them at least partially. The data dictionary is driven from a business perspective, and prioritises business-oriented definitions. Planning for new collections can become a straightforward exercise in data composition, readily informed by existing practice in other collections. The pilot phase of the project seeks to establish that the data dictionary can be harnessed by non-technical users not only to plan for, but also to

update the requirements of data collections, without substantial reengineering. The granularity of the data dictionary is also a critical enabler for agile updating of reports, as reporting requirements change.

- It aligns **data collections to a centralised, granular repertoire of data element definitions**. This means that the profusion of data collections can be managed more effectively, and it can reduce duplicated effort in gathering the same information. Efficient management of data reporting is becoming a pressing need, with an increasing priority on evidence-based decision making; greater expectation of information sharing between agencies; and more data becoming available for reporting, in divergent formats.
- It can serve as a more robust basis for **enforcing privacy requirements and data sharing policies**. The decisions on what information elements are sensitive, and how sensitive they are, involve business definitions of information, and are not dependent on a specific technical model. The data dictionary can map semantically defined sensitive fields to various data formats, without hard-coding to a specific data model. The data dictionary can point to indicative privacy ratings for each element in various collections and school authorities.
- It can serve as a mechanism for **communicating to the market about policy driven change to data and data exchange requirements**, for both existing established processes (e.g. the ACARA Data Standards Manual), and emerging process (wellbeing, child safety, formative assessment, learner mobility).